

Replicate Antikythera - User Manual

Applied Programming Concepts (ELEC 3225)

Professor Rawlins

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Table of Contents:

Section I: Introduction

- ❖ I.a: General Introduction
- ❖ I.b: Initial Setup

Section II: Running the Program

Section III: Interacting with the GUI

- ❖ III.a: Draw System Function
- ❖ III.b: Date Selection
- ❖ III.c: Major and Minor Bodies Menu
- ❖ III.d: Show Eclipses Menu

Section IV: Other Functions/Program Highlights

- ❖ IV.a: Alternate Way to Exit Program

Section I: Introduction

Section I.a: General Introduction:

Welcome to the Replicate Antikythera User Manual.

This guide will explain to you how to use this program's features, which include:

- Seeing where planets will be located in the solar system on a particular date
- Planet (major body) information based on planet that is chosen
- Comet & Asteroid (small body) information based on body that is chosen
- Information on Earth's Moon
- Highlight any lunar/solar eclipses that occurred on a particular date

Before you begin, please read the section below on the setup you must complete prior to running this code:

Section I.b: Initial Setup:

Please see the instructions on how to get the program to run properly below:

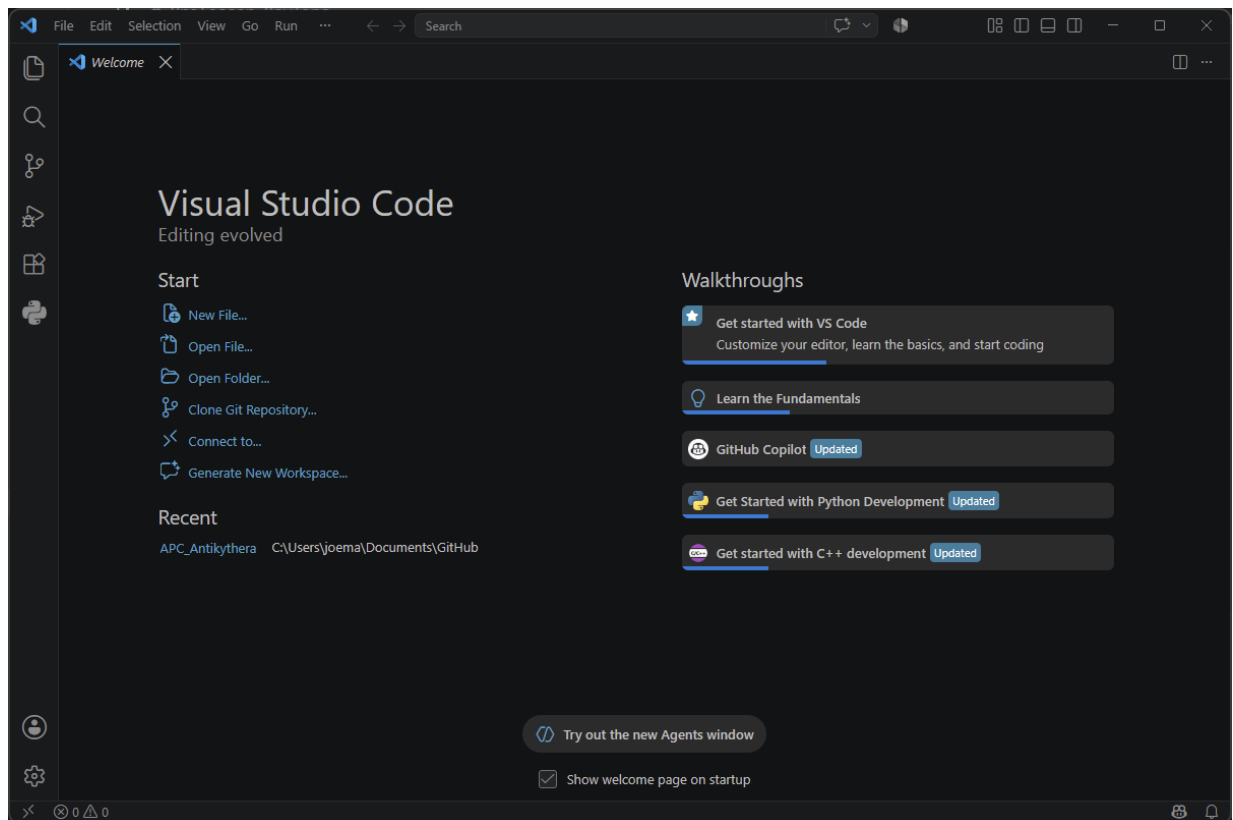
*Note: in order for this project to run properly on a Windows 11 PC, you **MUST** have Python version 3.10 (or higher) installed locally on your computer. A link to version 3.14.7 is [here](#). Please download the windows installer (MSIX) and go through the installation process. When the CMD window closes, you may proceed to the steps below*

1. Open the github link ([here](#))
2. Press the green "<> Code" button and look for a "download ZIP" button located at the bottom of the menu. This will now download the repo to your computer.
 - a. **DO NOT** place this zip file in your OneDrive, or any locations associated with your OneDrive (Desktop, Documents, Photos, etc). Place the zip file either in your downloads folder, or in your root drive (C:\). Visual Studio will refuse to open the file if it's in a OneDrive location as it doesn't have the proper permissions to open it.
3. From the location you saved the zip folder to, extract the zip file. You should see a folder structure very similar to this inside the folder called *APC_Antikythera*

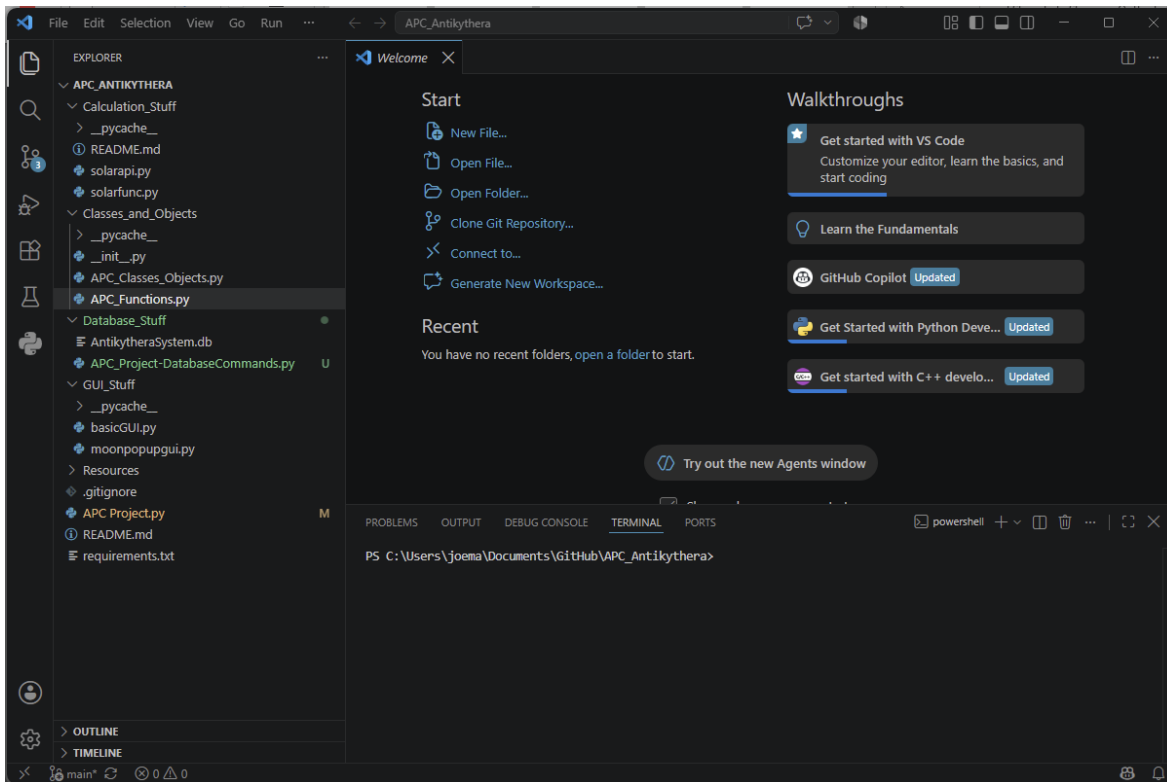
Calculation_Stuff	8/6/2026 8:37 AM	File folder	
Classes_and_Objects	8/6/2026 8:38 AM	File folder	
Database_Stuff	8/6/2026 8:30 AM	File folder	
GUI_Stuff	8/6/2026 8:24 AM	File folder	
Resources	7/19/2026 10:52 AM	File folder	
.gitignore	8/1/2026 10:21 AM	Git Ignore Source File	1 KB
APC Project.py	8/6/2026 8:43 AM	Python source file	1 KB
README.md	8/1/2026 10:21 AM	MD File	6 KB
requirements.txt	8/1/2026 10:21 AM	TXT File	1 KB

You'll want to point Visual Studio Code at the *APC Antikythera* folder. Let's do this now:

4. Open Visual Studio Code, and press: File -> New Window. You should see something that looks like this:



5. From here, you're going to press "Open Folder", and then point VS Code to that extracted *APC_Antikythera* folder on your PC.
6. After that, your VS Code window should look something like this:



If you see on the left-hand side of your screen an *Explorer* section with some folders, you can move onto the next step. If you don't then you pointed the wrong folder to VS Code.

7. Before you can run the program, you need to install some requirements in order to make the system work.

To do this, you will need to click into the *Terminal* tab on the bottom of the screen. When you are there, you should see a CMD line that starts with: *PS*

C:\Users\YOUR_USERNAME\FILE_LOCATION_PROJECT_IS_STORED\APC_Antikythera>

8. Press right next to the ">" and type in the following command: **py -m pip install -r requirements.txt**

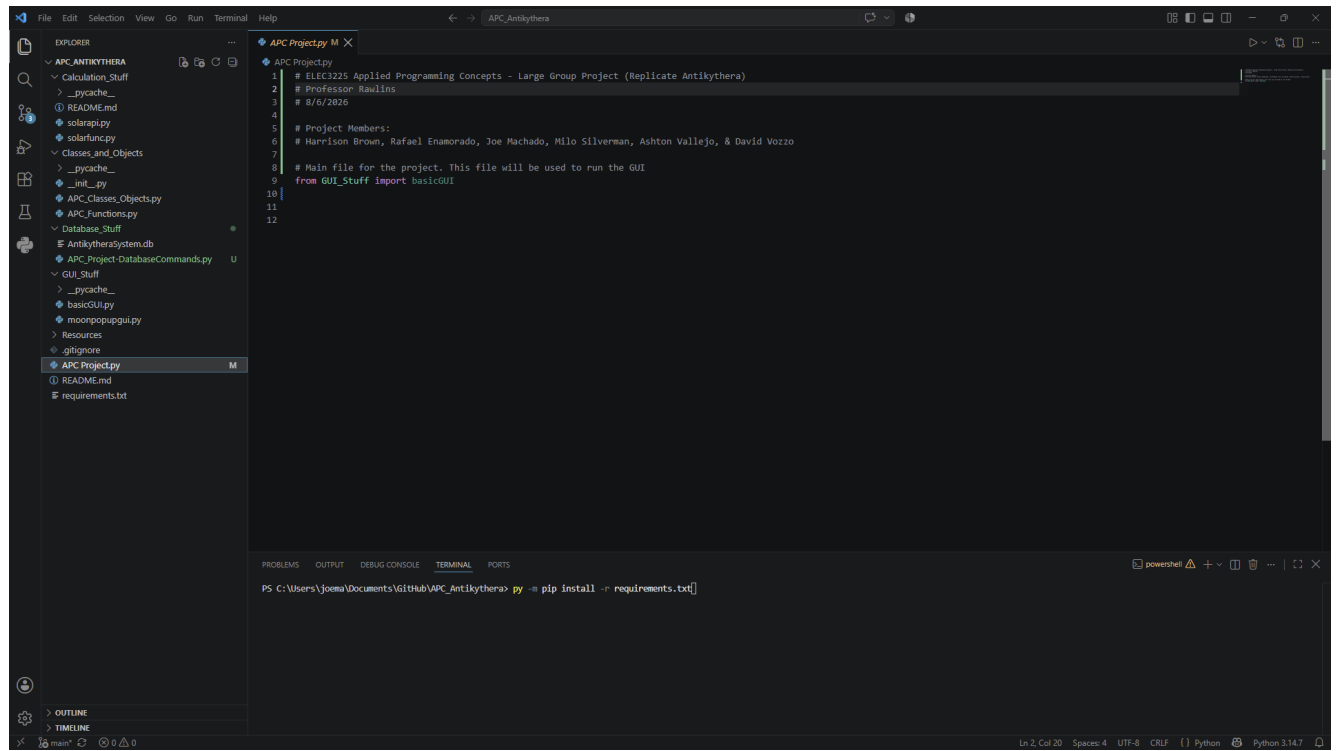
- a. This will install all the dependencies that this program needs to run properly.

- b. If the above command gave you an error, try: **python -m pip install -r requirements.txt**

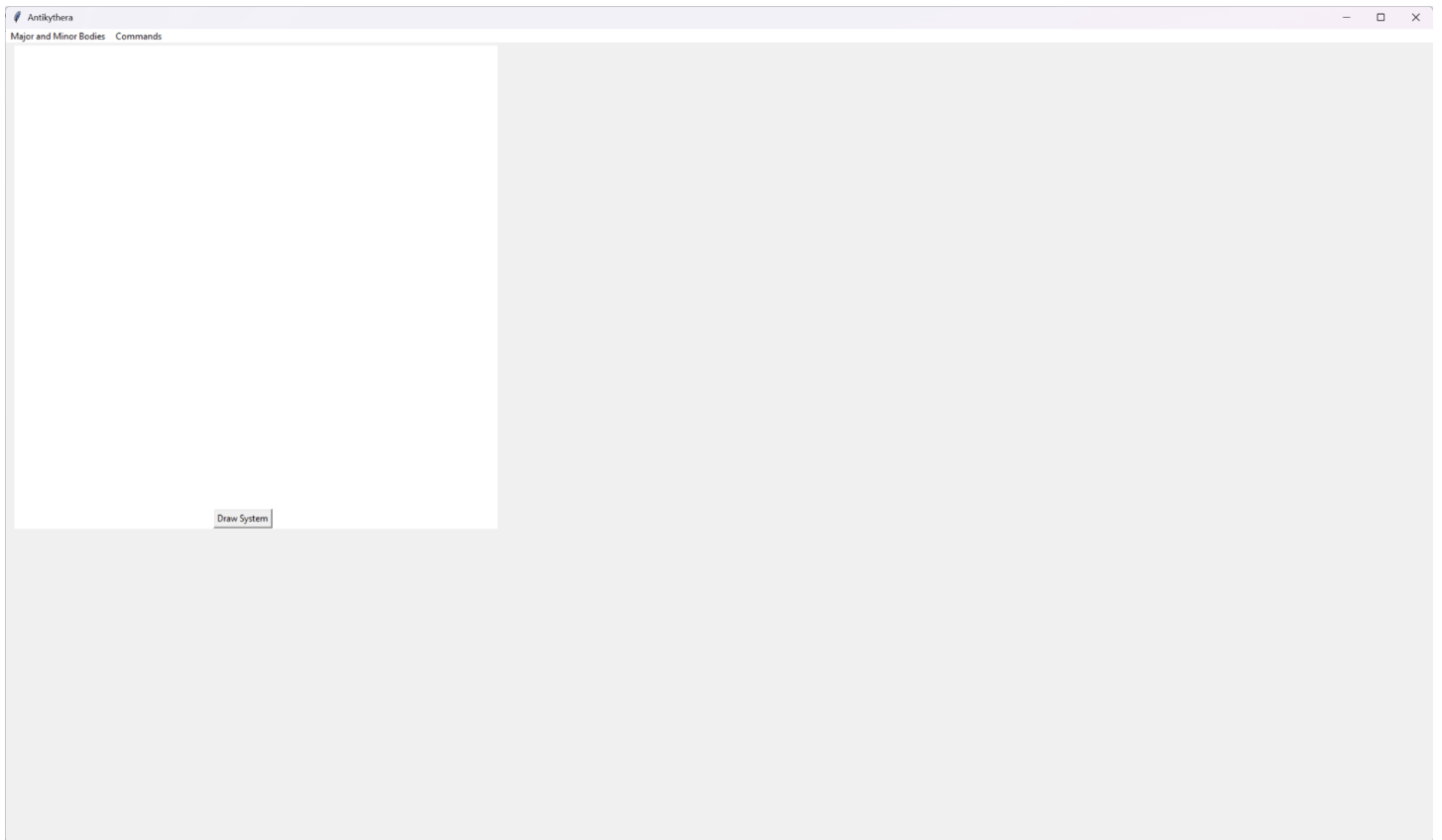
9. From here, you should see 4 libraries start to be installed. When they are done, you should see the CMD line from step 7 appear again. At this point, you are ready to run the program!

Section II: Running the Program

In the Explorer window on the left hand side, look for a file called *APC Project.py*. Double click it and you should see something that looks like this:



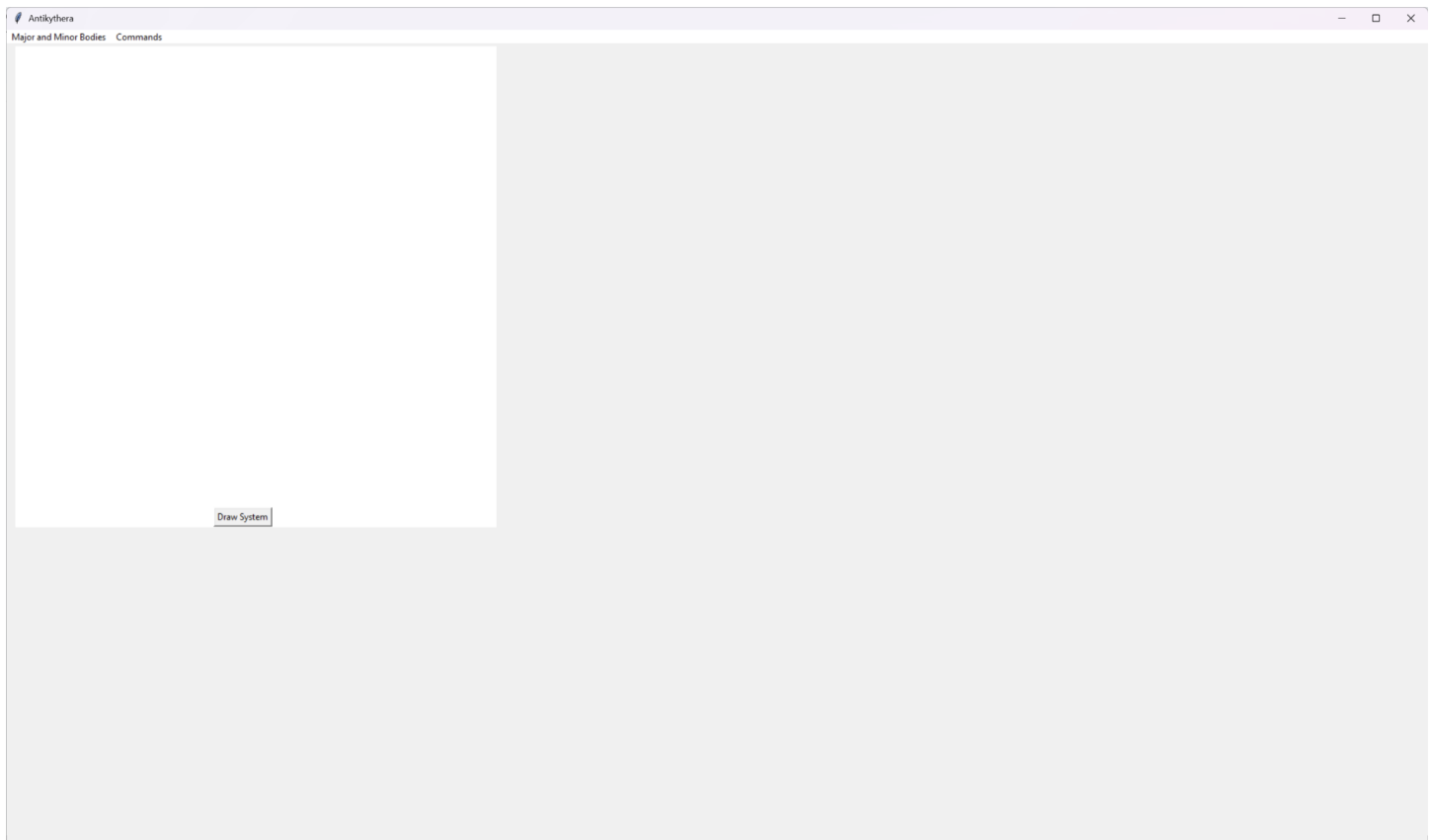
From this point, push the Play button at the top right of the screen and you should see a GUI appear that looks something like this:



See below for information about what you can do in the program:

Section III: Interacting with the GUI

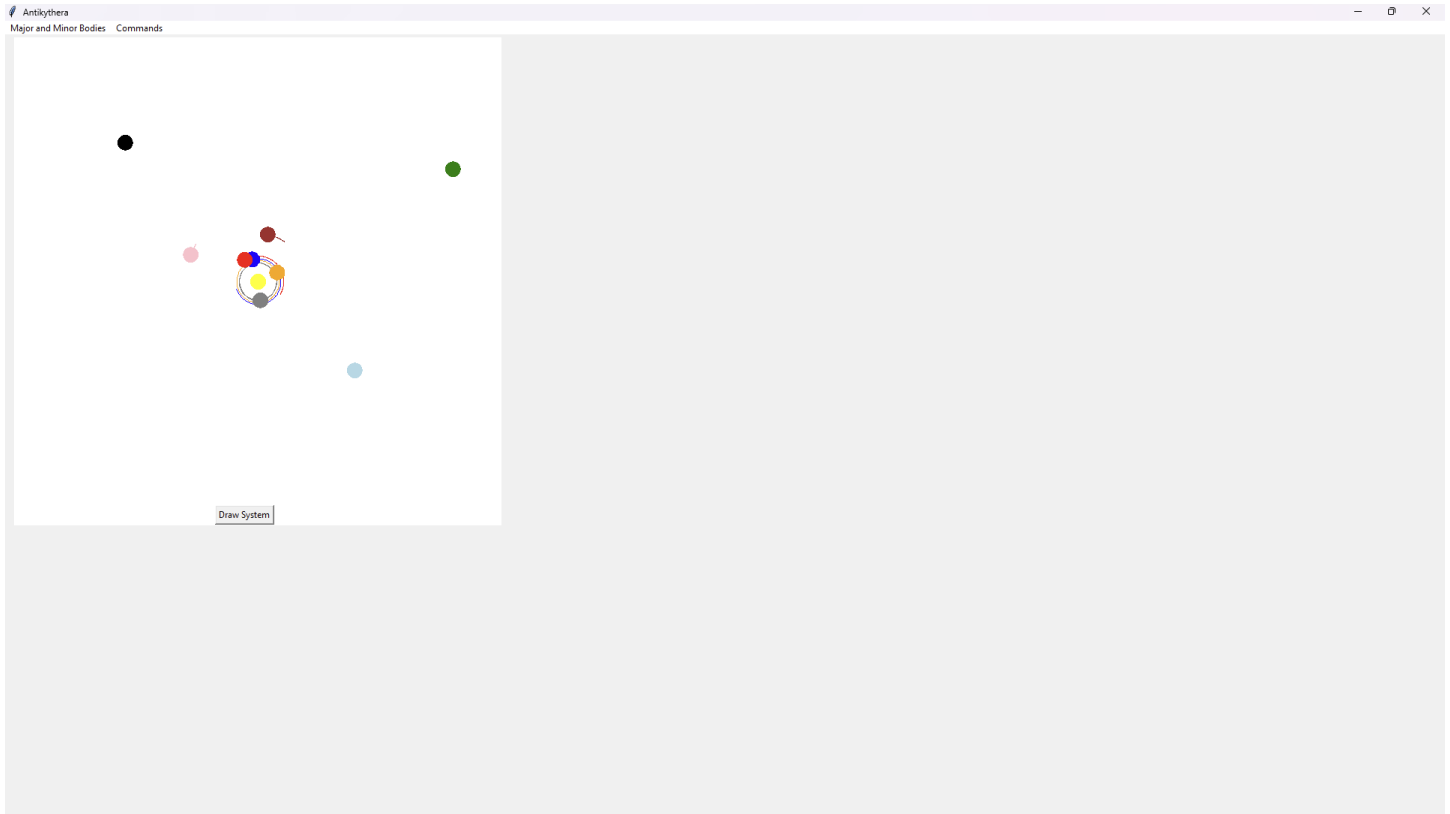
As shown above, when you first open the program, you will see a window that looks like this:



This is the main window that you will interact with and where all the menus will show up in (when you open them). You will first see a white box with a button below it labeled “Draw System”. This will be where a solar system model will show up with an animation of the planets revolving around the sun.

The only other thing on here that is worth noting is the menu bar. There are two main menus that are here: the “Major and Minor Bodies” which allows you to choose what astronomical bodies you want to see the information on. The other menu is called “Commands”, and in that menu it allows you to bring up the date selection as well as an alternate way to quit the program.

Section III.a: Draw System Function

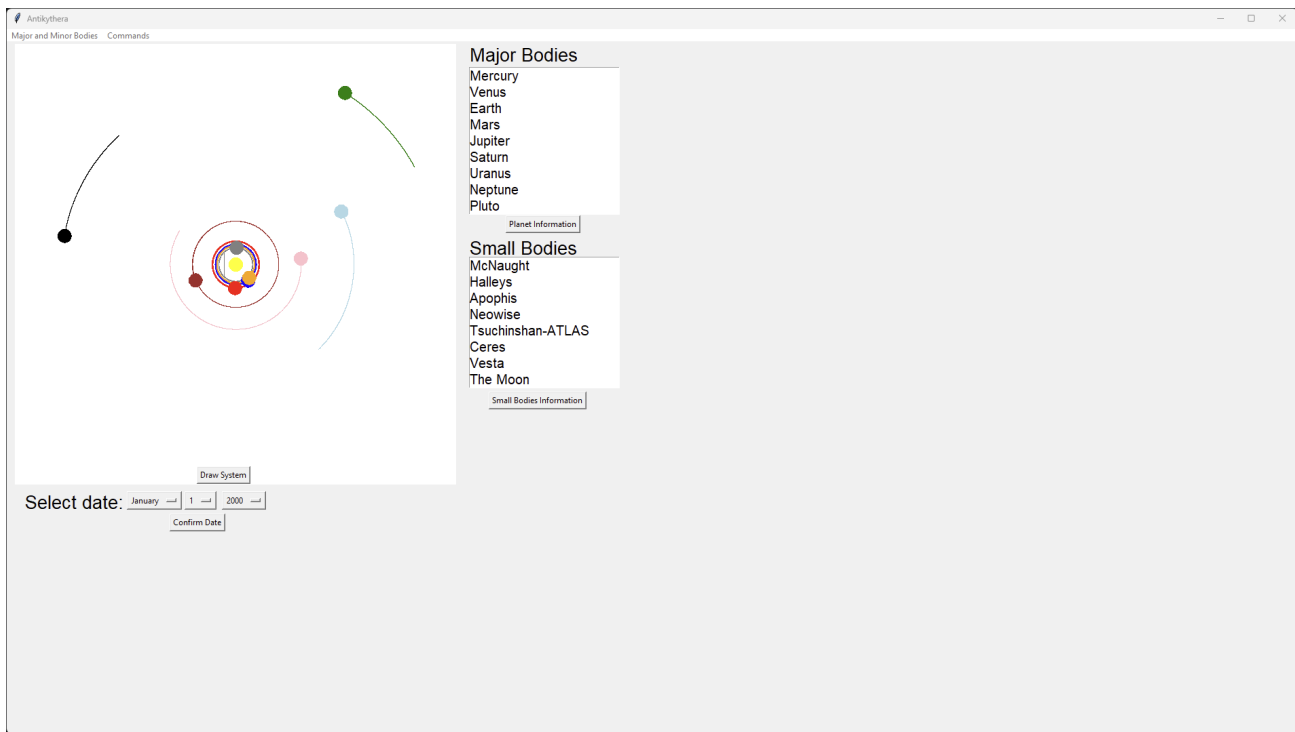


When you press the “Draw System” button, it will do a small animation of all the planets coming out from the center and then start revolving around a yellow circle (which is meant to be the Sun). This function will continuously loop until you exit the program.

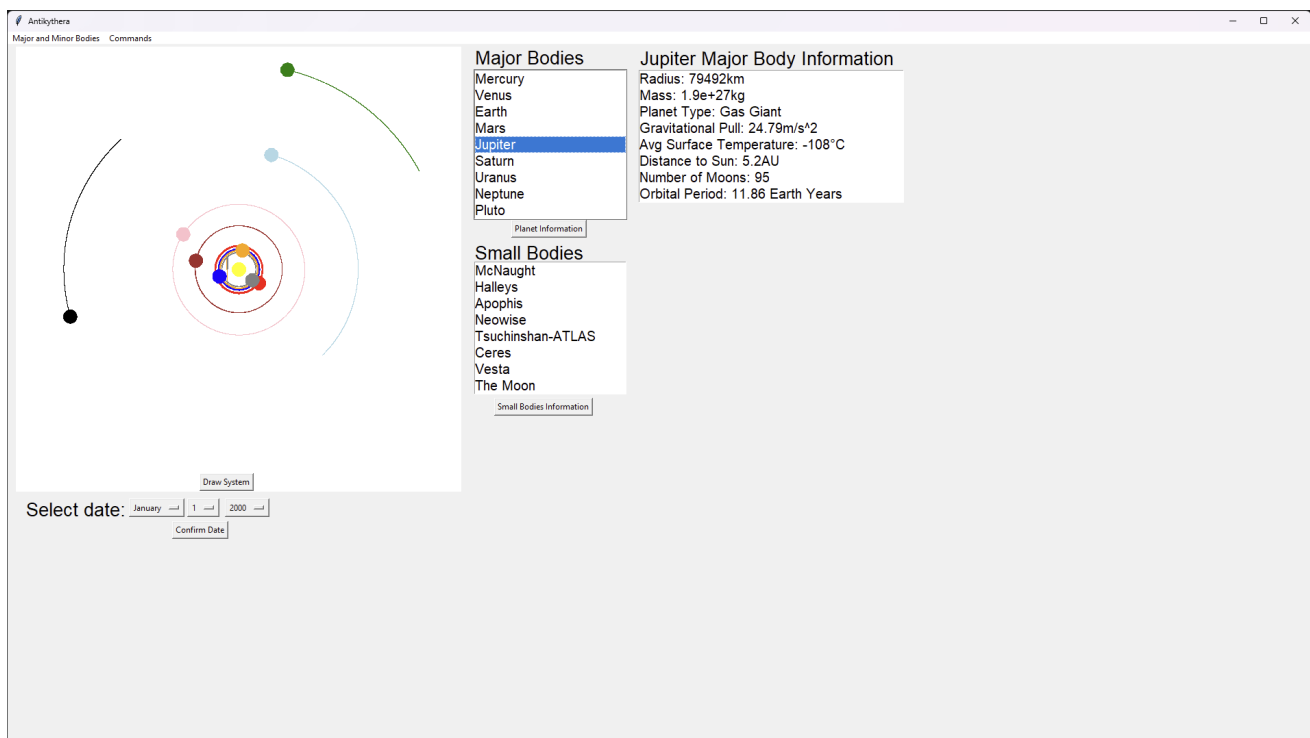
Section III.b: Major and Minor Bodies Menu

When you first open the “Major and Minor Bodies” menu, you’ll see two options, one for “Open Planet Selection” and one for “Open Comet and Asteroid Selection”. When you press the *Planet Selection* menu, you will see a small selection window that will pop up with all the planets and a button that says “Planet Information”. Similarly, when you press the *Comet and Asteroid Selection* menu, you will see a selection menu for all the comets and asteroids as well as a button similar to the one above labeled *Small Bodies Information*,

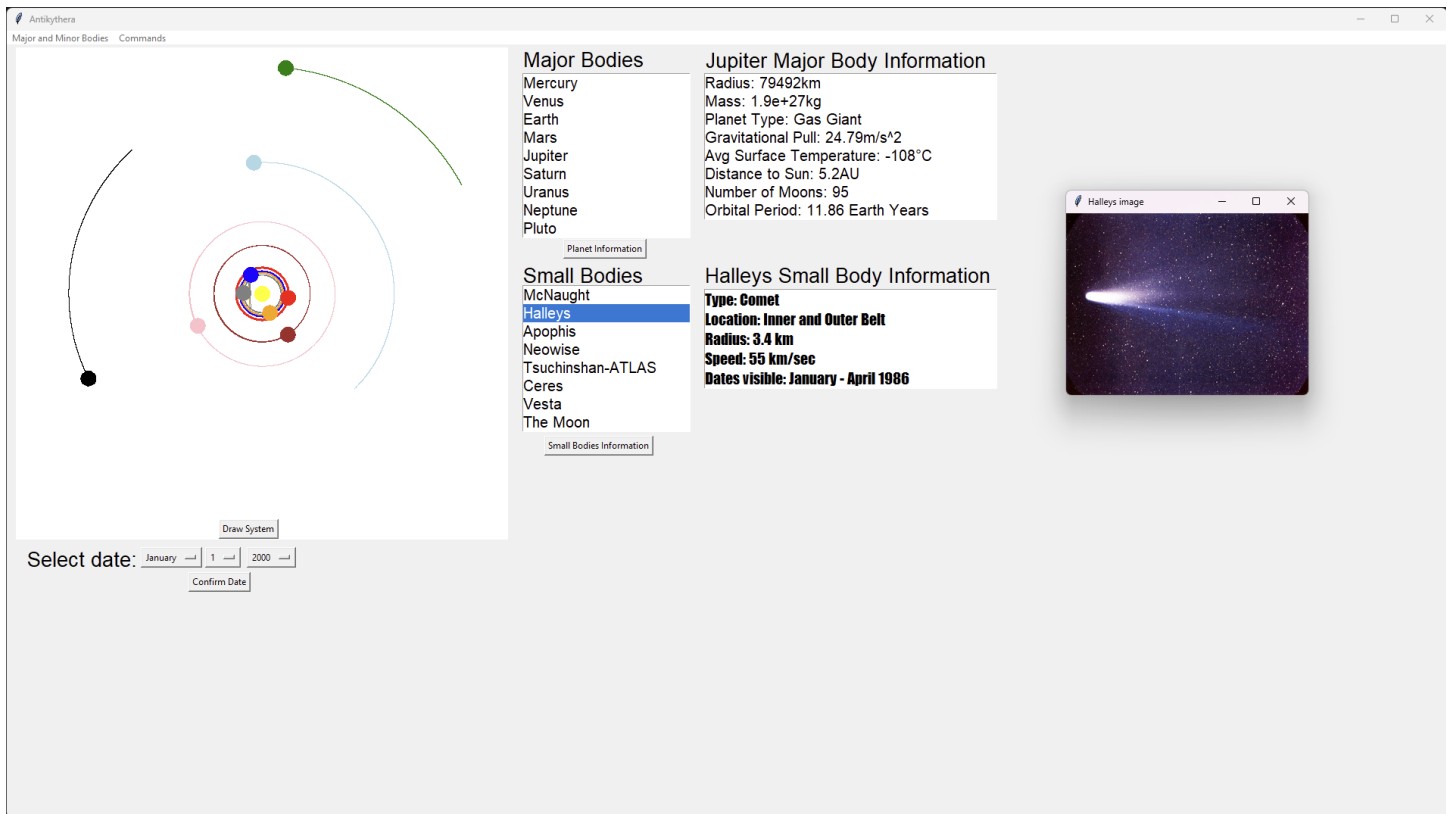
When both menus are open at the same time your screen will look something like this:



If you want to see the information on a certain planet, you can press the *Planet Information* button while a planet is selected. An example for the planet *Jupiter* is below:



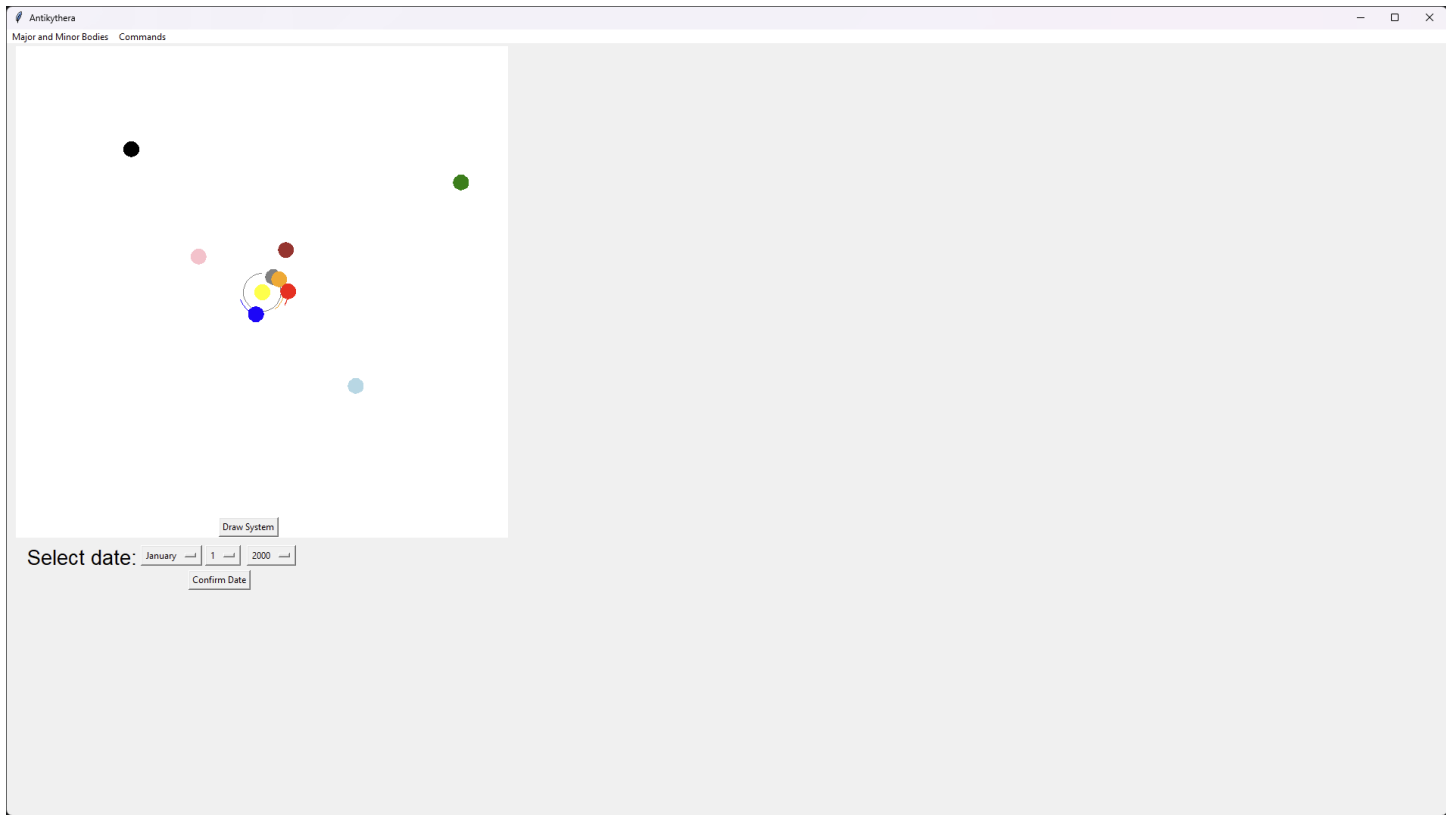
This is the same idea for Comets and Asteroids, but a photo of the selected comet/asteroid is also shown. An example for the comet *Haileys* is shown below:



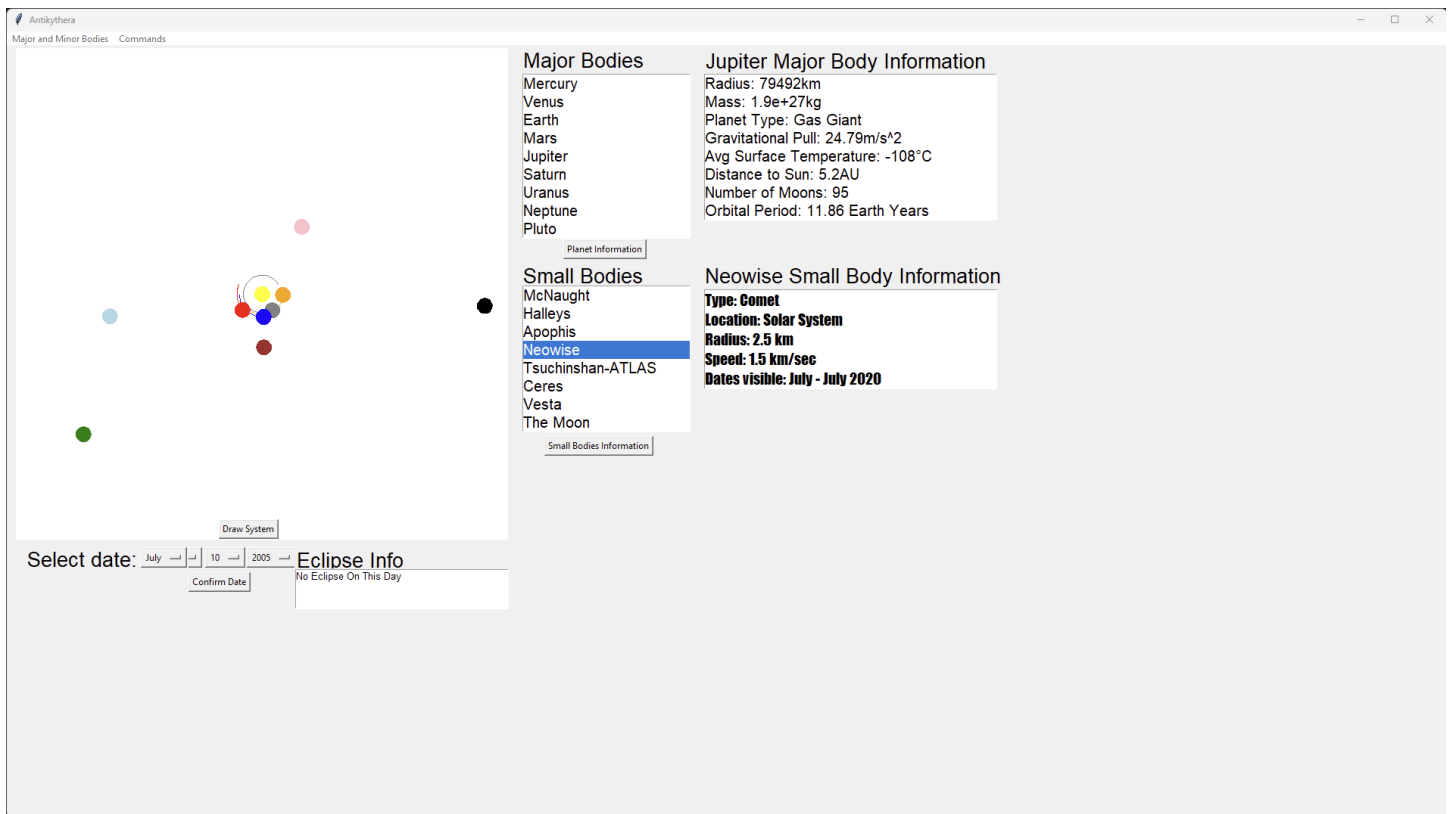
Section III.c: Commands Menu

The *Commands* menu doesn't really have much to it. The only thing that is worth noting in this menu is the "Open Date Selection" command. This brings up a small date selection window on the right side of the solar system block. All the selection options here are dropdown menus, so when you select your options, just use the dropdown menu to select your date. When you enter a date and press *Confirm Date*, there will be another menu that shows up which will contain information about any Eclipses that occurred on that date. This will be explained more in-depth below.

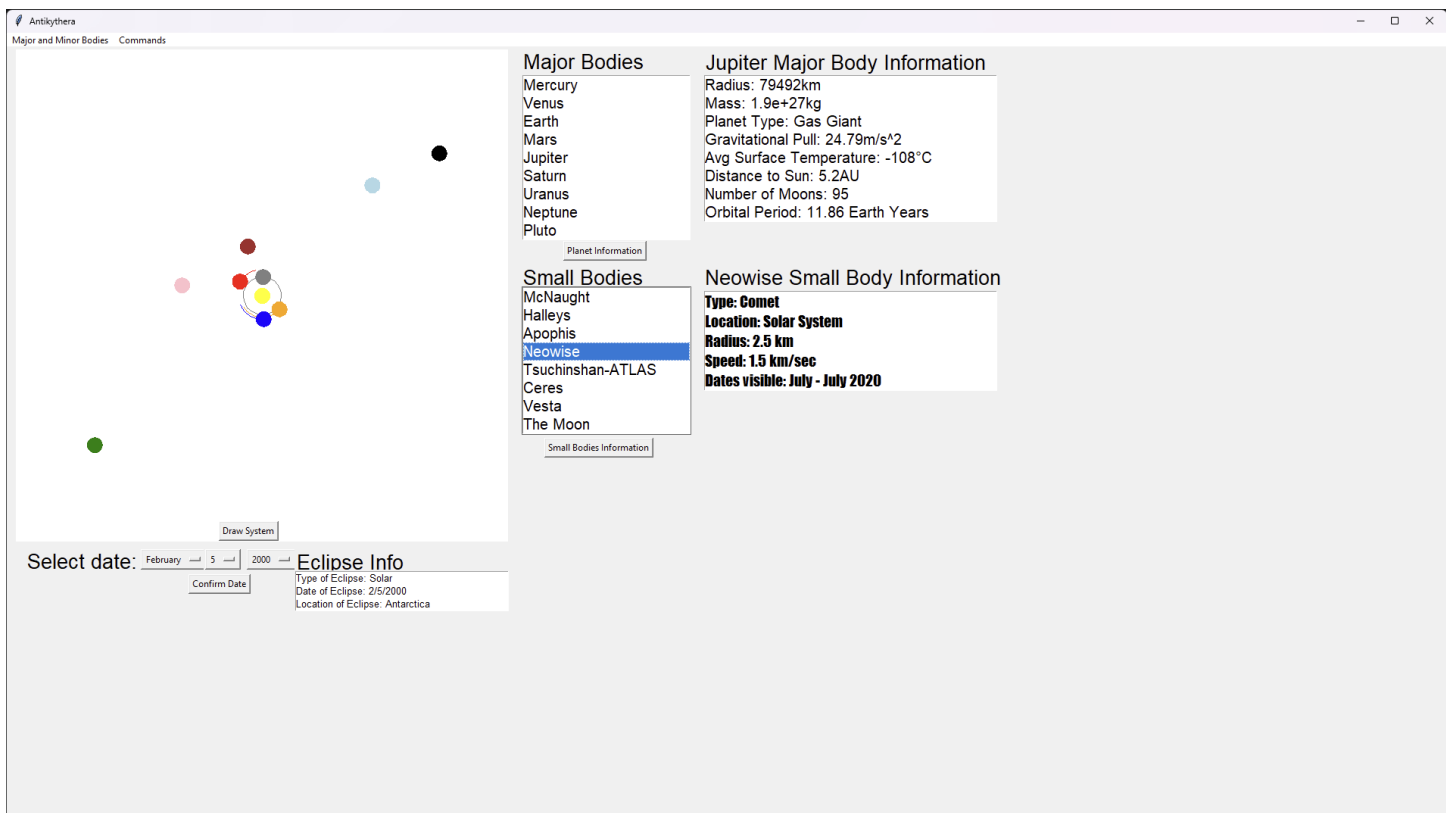
The date selection window will look something like this:



When pressing the *Confirm Date* button, the planets in the solar system will get redrawn based on the date you entered. An example of the system being re-drawn for the date *July 10th, 2005* is shown below.



As you can see, there is also a small window next to the date selection area that shows if any eclipses appeared on that day. In the photo above, there is no eclipse on that day but in the photo below, there is a solar eclipse.



Section IV: Other Functions/Program Highlights

Section IV.a Alternate Way to Exit Program

If for some reason you don't want to use the "X" in the corner of the window to quit, you can also press the *Exit Program* button in the *Commands* menu to exit the program.

The photo below shows the *Exit Program* button highlighted:

